

MACHINE LEARNING CASE STUDY

Bank Term Deposit **Prediction**

Comparing Logistic Regression & Decision Tree Classifiers for Direct Marketing Optimization

Executive Summary

This project builds an end-to-end classification pipeline to predict customer conversion for bank term deposit subscriptions, optimizing telemarketing efficiency and maximizing marketing ROI.

Data Preprocessing Pipeline



ColumnTransformer

Numerical features are processed with **StandardScaler**, while categorical variables are converted using **OneHotEncoder** for seamless modeling.



Leakage Prevention

The preprocessor fits **strictly on training data (x_tr)**. Validation (x_val) and test sets (x_test) are transformed using learned training statistics.



Outlier Preservation

Standard scaling keeps extreme values linear (Z-scores), ensuring robust inputs for Logistic Regression while Decision Trees remain scale-invariant.

Model Performance Comparison

Model Split	Accuracy	Precision	Recall	F1-Score	Status
Logistic Regression (Validation)	83.65%	82.74%	82.74%	82.74%	Baseline
Logistic Regression (Test)	82.40%	82.13%	80.34%	81.22%	W Winner
Decision Tree (Tuned Test)	80.79%	77.52%	83.74%	80.51%	Candidate

Logistic Regression demonstrates strong generalization on unseen test data with minimal drop (~1.2%), avoiding overfitting while maintaining higher precision.

Test Performance Benchmark



Logistic Regression achieves higher Precision (82.1% vs 77.5%) and overall Accuracy, making it optimal for reducing false positive call costs.

Model Mechanics & Trade-offs

Logistic Regression

Strengths: High precision (82.13%) reduces wasted telemarketing effort. Provides direct probability estimation and clear coefficient interpretability for business decisions.

Trade-off: Slightly lower recall (80.34%) compared to decision trees.

Decision Tree Classifier

Strengths: Higher recall (83.74%) captures a broader proportion of potential subscribers without needing numerical scaling.

Trade-off: Lower precision (77.52%) leads to more false alarms and elevated contact costs.

Key Positive Drivers

- ✓ **Previous Success (poutcome +2.26):** Past successful conversion is the single strongest predictor of subscription.
- ✓ **Call Duration (+1.93):** Longer customer engagement directly correlates with higher subscription likelihood.
- ✓ **Off-Peak Months:** Calls made in March (+1.67), December (+1.04), and October (+1.00) yield top conversion rates.
- ✓ **Higher Education (+0.53):** Clients with tertiary education show greater inclination to open term deposits.

Strongest Negative Predictors

- ✗ **Unknown Contact Channel (-1.65):** Unrecorded contact methods drastically decrease campaign success rates.
- ✗ **Low-Converting Months:** January (-1.26), November (-0.98), and July (-0.98) show significant negative conversion odds.
- ✗ **Existing Liabilities:** Active housing mortgages (-0.74) and personal loans (-0.51) reduce disposable liquidity.

Strategic Recommendations

-  **Prioritize Past Responders:** Direct sales teams toward leads with a positive prior campaign record (poutcome == 'success').
-  **Fix Data Hygiene:** Standardize communication tracking to eliminate 'unknown' contact channels across sales systems.
-  **Optimize Seasonality:** Shift aggressive outreach to high-performing months (March, September, October, December).
-  **Agent Training on Engagement:** Focus agent script training on building call duration, as call length strongly drives conversion.

Winning Model: Logistic Regression

Selected for deployment due to superior Precision (82.13%), balanced F1-score (81.22%), and optimal return on telemarketing investment.

82.40%

Accuracy

82.13%

Precision

81.22%

F1-Score